Machine Learning for social sciences

Natural Language Processing: Session 2

William Aboucaya

Natural Language Inference

Natural Language Inference (NLI) is "the problem of determining **entailment** and **contradiction** relationships between a **premise** and a **hypothesis**"[1]

[1] Ankur Parikh et al. "A Decomposable Attention Model for Natural Language Inference". In: *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. Texas: Association for Computational Linguistics, Nov. 2016.

Natural Language Inference

Natural Language Inference (NLI) is "the problem of determining **entailment** and **contradiction** relationships between a **premise** and a **hypothesis**"[1]

$\Rightarrow$ Does the **hypothesis** logically follow from the **premise**?

[1] Ankur Parikh et al. "A Decomposable Attention Model for Natural Language Inference". In: *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. Texas: Association for Computational Linguistics, Nov. 2016.

Natural Language Inference

Natural Language Inference (NLI) is "the problem of determining **entailment** and **contradiction** relationships between a **premise** and a **hypothesis**"[1]

⇒ Does the **hypothesis** logically follow from the **premise**?

Labels:

- **Entailment:** Hypothesis is true if the premise is true.
- **Contradiction:** Hypothesis is false if the premise is true.
- **Neutral:** Hypothesis is neither entailed nor contradicted by the premise.

[1] Ankur Parikh et al. "A Decomposable Attention Model for Natural Language Inference". In: *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. Texas: Association for Computational Linguistics, Nov. 2016.

Natural Language Inference

Applications of NLI

- Question Answering Systems
- Information Retrieval and Summarization
- Sentiment Analysis and Opinion Mining
- Verifying Claims in Automated Fact-Checking

Natural Language Inference

Example:

- **Premise:** My cat is in the tree
- **Hypothesis:** There is at least one cat in the tree
- **Scores:**
  - **Entailment: 0.98**
  - Contradiction: 0.00
  - Neutral: 0.02

Results obtained using `ynie/roberta-large-snli_mnli_fever_anli_R1_R2_R3-nli`

Natural Language Inference

Example:

- **Premise:** My cat is in the tree
- **Hypothesis:** My cat never left the house
- **Scores:**
  - Entailment: 0.00
  - **Contradiction: 0.93**
  - Neutral: 0.07

Results obtained using `ynie/roberta-large-snli_mnli_fever_anli_R1_R2_R3-nli`

Natural Language Inference

Example:

- **Premise:** My cat is in the tree
- **Hypothesis:** It is winter
- **Scores:**
  - Entailment: 0.01
  - Contradiction: 0.01
  - **Neutral: 0.98**

Results obtained using `ynie/roberta-large-snli_mnli_fever_anli_R1_R2_R3-nli`

NLI data

What do NLI datasets look like?

[2] Adina Williams, Nikita Nangia, and Samuel Bowman. "A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference". In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Louisiana: ACL, June 2018.

NLI data

What do NLI datasets look like?

Example: the Multi-genre Natural Language Inference Corpus (MNLI)[2]

- **A premise:** At the other end of Pennsylvania Avenue, people began to line up for a White House tour.
- **A hypothesis:** People formed a line at the end of Pennsylvania Avenue.
- **A label:** Entailment

⇒ These datasets are used to fine-tune and evaluate language models

[2] Adina Williams, Nikita Nangia, and Samuel Bowman. "A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference". In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Louisiana: ACL, June 2018.

Challenges in NLI

- Handling ambiguity and polysemy
- Dealing with implicit knowledge and commonsense reasoning
- Managing syntactic and semantic variability
- Training robust models on diverse datasets (models tend to measure similarity rather than entailment[3])

[3] Adina Williams, Nikita Nangia, and Samuel Bowman. "A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference". In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Louisiana: ACL, June 2018.

Challenges in NLI

- Handling ambiguity and polysemy
- Dealing with implicit knowledge and commonsense reasoning
- Managing syntactic and semantic variability
- Training robust models on diverse datasets (models tend to measure similarity rather than entailment[3])

Let's see what we can do with it 🤓

[3] Adina Williams, Nikita Nangia, and Samuel Bowman. "A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference". In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Louisiana: ACL, June 2018.

Zero-shot learning

Zero-shot learning (ZSL) is a machine learning paradigm where a model can make predictions for classes or tasks it has never explicitly seen during training using **knowledge transfer**.

Zero-shot learning

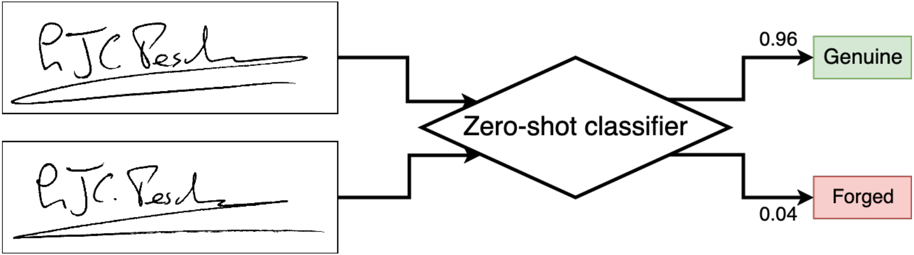
Zero-shot learning (ZSL) is a machine learning paradigm where a model can make predictions for classes or tasks it has never explicitly seen during training using **knowledge transfer**.

Key Components of ZSL:

- **Seen Classes:** Classes/tasks available during training.
- **Unseen Classes:** Classes/tasks not available during training.
- **Knowledge Representation:** Attributes or embeddings that describe both seen and unseen classes.
- **Model Objective:** Learn a mapping between input features and the shared knowledge space.

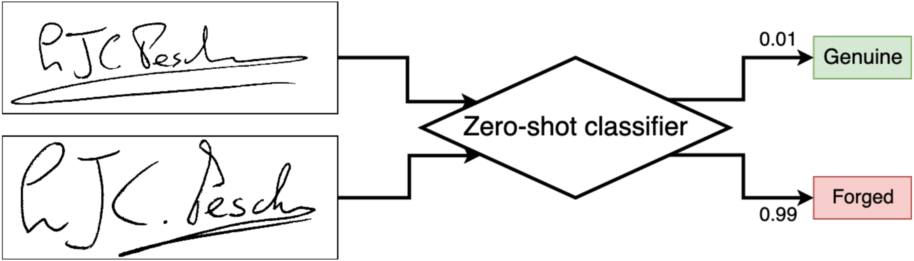
Zero-shot learning

Example: Signature genuineness evaluation



Zero-shot learning

Example: Signature genuineness evaluation



Zero-shot classification

What is it?

- A NLP task where a model classifies documents into categories **without explicit training** on those categories
- Leverages **pre-trained language models** based on **prompting** or **NLI**

Zero-shot classification

What is it?

- A NLP task where a model classifies documents into categories **without explicit training** on those categories
- Leverages **pre-trained language models** based on **prompting** or **NLI**

Advantages:

- **No Task-Specific Data Required**: Works directly with new labels
- **Rapid Prototyping**: Useful for tasks with little or no annotated data

Zero-shot classification

Applications:

- Content moderation
- Sentiment analysis
- Topic classification
- Legal or medical text categorization
- etc.

Zero-shot classification

Applications:

- Content moderation
- Sentiment analysis
- Topic classification
- Legal or medical text categorization
- etc.

Example:

- **Input:** *"European Parliament adopts 'historic' AI Act"*
- **Labels:** `Technology, Sports, Politics`
- **Predictions:** `Technology, Politics`

Results obtained using `MoritzLaurer/deberta-v3-large-zeroshot-v2.0`

How to do it?

- A first approach: Just ask ChatGPT 😶

Zero-shot classification

How to do it?

- A first approach: Just ask ChatGPT 😶
- No, really, the results obtained by asking GPT-4, Llama 3 or MISTRAL are... fine[4]

[4] Martin Juan José Bucher and Marco Martini. *Fine-Tuned 'Small' LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification*. 2024. arXiv: 2406.08660 [cs.CL].

Zero-shot classification

How to do it?

- A first approach: Just ask ChatGPT 😶
- No, really, the results obtained by asking GPT-4, Llama 3 or MISTRAL are... fine[4]

Issues of this approach:

- **Scalability:** The larger the model, the longer it takes to compute each classification
- **Price:** If you want to label hundreds of thousands of documents, asking a remote service to produce millions of tokens isn't cheap
- **Reliability:** Those models do not produce confidence intervals for the labels

[4] Martin Juan José Bucher and Marco Martini. *Fine-Tuned 'Small' LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification*. 2024. arXiv: 2406.08660 [cs.CL].

Zero-shot classification

An approach to zero-shot classification based on NLI

- Define a set of candidate labels.
- Convert each label into a hypothesis (e.g., "This text is about *sports*.").
- Use an NLI model to compute entailment scores: Higher score $\rightarrow$ Higher confidence in the relevance of the label
- Assign the label with the highest entailment score or a list of labels with a score higher than a threshold

Zero-shot classification

An approach to zero-shot classification based on NLI

- Define a set of candidate labels.
- Convert each label into a hypothesis (e.g., "This text is about *sports*.").
- Use an NLI model to compute entailment scores: Higher score $\rightarrow$ Higher confidence in the relevance of the label
- Assign the label with the highest entailment score or a list of labels with a score higher than a threshold

Let's see what we can do with it 🤓